2 Estimating and Evaluating Asset Pricing Models

Regression-Based Tests of Linear Factor Models

2.1 Time-Series Regressions [TSR]

- Let's test a single-factor asset-pricing model:

$$E(r_i) = \beta_i E(f) \tag{33}$$

where r_i are asset excess returns and f are the excess returns on a factor.

If the factor is an excess return, we can evaluate the model *with time-series OLS regressions* of the form:

$$r_{i,t} = \alpha_i + \beta_i f_t + \epsilon_{i,t}. \tag{34}$$

- The asset pricing model says that the regression intercepts must be zero.
- **The regression intercepts, α_i, reflect the pricing errors as long as the factors are return spreads.**

- So we would like to
  test whether all the pricing errors are jointly equal to zero:

$$\mathbf{H_0} : \alpha \equiv [\alpha_1, \alpha_2, ..., \alpha_N]' = \mathbf{0}. \tag{35}$$

- Next we will derive a test statistic to test (35).
- We will use concepts of maximum likelihood [ML] estimation.
- Campbell, Lo, and MacKinlay (1997, ch.5) derive the result with ML, while Cochrane (2001, ch.12) derives it with GMM. They both reach the same result.

2.1.1 An asymptotic test statistic

- To test whether all the pricing errors are jointly zero using ML, we assume that the regression residuals are IID Normal (GMM relaxes this assumption).
- The ML principle says to pick the parameters that make the data most likely.

1. Rewriting equation (34) in its multivariate form yields

$$r_t = \alpha + \beta f_t + \epsilon_t. \tag{36}$$

where $r_t \equiv [r_{1,t}, r_{2,t}, ..., r_{N,t}]'$, $\beta \equiv [\beta_1, \beta_2, ..., \beta_N]'$, $\epsilon_t \equiv [\epsilon_{1,t}, \epsilon_{2,t}, ..., \epsilon_{N,t}]'$ for $t = [1, 2, ..., T]$.

2. Under normality, the likelihood function for ϵ_t is

$$\mathcal{L}(\epsilon_t|\alpha,\beta,\Sigma) = (2\pi)^{-\frac{N}{2}}\Sigma^{-\frac{1}{2}}\exp\left[-\frac{1}{2}(r_t-\alpha-\beta f_t)'\Sigma^{-1}(r_t-\alpha-\beta f_t)\right]$$

Moreover, the IID assumption suggests that for T observations

$$\mathcal{L}(\epsilon_1,\epsilon_2,\ldots,\epsilon_T|\alpha,\beta,\Sigma) = (2\pi)^{-\frac{TN}{2}}\Sigma^{-\frac{T}{2}} \tag{37}$$
$$\times\exp\left[-\frac{1}{2}\sum_{t=1}^{T}(r_t-\alpha-\beta f_t)'\Sigma^{-1}(r_t-\alpha-\beta f_t)\right]$$

3. The maximum likelihood estimators [MLEs] are the value of the parameters which maximize $\mathcal{L}$ in (37). To find these estimators, we differentiate $\mathcal{L}$ with respect to α, β, and Σ and and set the derivatives equal to zero.

$$\frac{\partial\mathcal{L}}{\partial\alpha} = \Sigma^{-1}\left[\sum_{t=1}^{T}(r_t-\alpha-\beta f_t)\right], \tag{38}$$

$$\frac{\partial\mathcal{L}}{\partial\beta} = \Sigma^{-1}\left[\sum_{t=1}^{T}(r_t-\alpha-\beta f_t)f_t\right], \tag{39}$$

$$\frac{\partial\mathcal{L}}{\partial\Sigma} = -\frac{T}{2}\Sigma^{-1}+\frac{1}{2}\Sigma^{-1}\left[\sum_{t=1}^{T}\epsilon_t\epsilon_t'\right]\Sigma^{-1} \tag{40}$$

Estimates:

4. The MLEs are given by

$$\hat{\alpha} = \hat{\mu} - \hat{\beta}\hat{E}(f), \tag{41}$$

$$\hat{\beta} = \frac{\sum_{t=1}^{T}(r_t - \hat{\mu})(f_t - \hat{E}(f))}{\sum_{t=1}^{T}(f_t - \hat{E}(f))^2}, \tag{42}$$

$$\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T}\hat{\epsilon}_t\hat{\epsilon}_t', \tag{43}$$

where $\hat{\mu} \equiv \frac{1}{T}\sum_{t=1}^{T} r_t$ and $\hat{E}(f) \equiv \frac{1}{T}\sum_{t=1}^{T} f_t \equiv \overline{f}$.
Note that these are just the OLS estimators of the parameters.

5. The estimate for the factor risk premium is the sample mean of the factor:

$$\hat{\lambda} = \hat{E}(f) \tag{44}$$

Standard Errors:

6. α_i and β_i have OLS standard errors (in each regression)
7. $\hat{\lambda}$

$$\sigma(\hat{\lambda}) = \frac{\sigma(f_t)}{\sqrt{T}} \tag{45}$$

8. **Let us now find the standard errors of $\hat{\alpha}$ to test whether α are *jointly zero***
9. We can form a Wald test statistic of the null hypothesis

$$\mathcal{H}_0 : \ \alpha = 0$$

against the alternative hypothesis

$$\mathcal{H}_1 : \ \alpha \neq 0.$$

10. The Wald statistic is

$$J_1 = \hat{\alpha}'[\text{Var}(\hat{\alpha})]^{-1}\hat{\alpha}. \tag{46}$$

11. We need the $\text{Var}(\hat{\alpha})$, which is affected by beta fitting.
12. Let us denote the set of estimates by $\hat{\Theta} = (\hat{\alpha}, \hat{\beta}, \hat{\Sigma})$.
    Asymptotically

$$\Theta - \hat{\Theta} \sim N(0, I(\Theta)^{-1}), \tag{47}$$

where $I(\Theta)$ is the **information matrix**—the matrix of second derivatives of the likelihood with respect to the parameters—given by

$$I(\Theta) = \begin{bmatrix} -T\Sigma^{-1} & -\Sigma^{-1}E(f) \\ -\Sigma^{-1}E(f) & -\Sigma^{-1}\left(E(f)^2 + \sigma(f)^2\right) \end{bmatrix}. \tag{48}$$

we disregard the third row and third column
since Σ is uncorrelated with α and β.

13. Thus, it follows that

$$\hat{\alpha} \sim N\left(\alpha, \frac{1}{T}\left[1 + \frac{\hat{E}(f)^2}{\hat{\sigma}(f)^2}\right]\Sigma\right). \tag{49}$$

14. **The $\mathbf{Var}(\hat{\alpha}) = \frac{1}{T}\Sigma$, except for beta fitting.**

    The $\left[1 + \left(\frac{\hat{E}(f)}{\hat{\sigma}(f)}\right)^2\right]$ term corrects for beta fitting.

15. Bottom line:

> The *time-series test statistic* for a single factor asset-pricing model is:
>
> $$J_1 = T\left[1 + \left(\frac{\hat{E}(f)}{\hat{\sigma}(f)}\right)^2\right]^{-1} \hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha} \quad \sim \quad \chi^2_N \tag{50}$$

- The *asymptotic* distribution of J_1 is χ^2_N.

2.1.2 A finite sample test

- A finite sample test statistic for the joint equality of the pricing errors to zero has been developed by Gibbons, Ross, and Shanken (1989) [GRS].

The GRS test statistic is

$$J_2 = \frac{T-N-1}{N}\left[1+\left(\frac{\hat{E}(f)}{\hat{\sigma}(f)}\right)^2\right]^{-1}\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha} \qquad \sim F_{N,T-N-1} \quad (51)$$

- The statistic J_2 has the *finite-sample* distribution $F_{N,T-N-1}$.
- **The algorithm for implementing the statistic is**:
  1. For each asset, run TS regressions of the asset returns on the factor.
  2. Retain the estimated regression intercepts $\hat{\alpha} = [\hat{\alpha}_1, \hat{\alpha}_2, \ldots, \hat{\alpha}_N]'$ and the estimated residuals $\epsilon_t = [\epsilon_t^1, \epsilon_t^2, \ldots, \epsilon_t^N]'$ for $t = [1, 2, \ldots, T]$.
  3. Compute the residual covariance matrix $\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T}\epsilon_t\epsilon_t'$.
  4. Compute the sample mean and the sample standard deviation of the factor $\hat{E}(f)$ and $\hat{\sigma}(f)$, respectively, and compute J_2 as in Equation 51.

- If there are **multiple factors**, the regression specification is

$$r_{i,t} = \alpha_i + \beta_i' f_t + \epsilon_{i,t}. \tag{52}$$

- The pricing restrictions are again: the pricing errors, α_i, should be equal to zero.

The GRS test statistic for multi-factor models is

$$J_3 = \frac{T-N-K}{N}\left[1+\hat{E}(f)'\hat{\Omega}^{-1}\hat{E}(f)\right]^{-1}\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha} \quad \sim \quad F_{N,T-N-K} \tag{53}$$

where K is the number of factors, $\hat{E}(f) = [\hat{E}(f_1), \hat{E}(f_2), ..., \hat{E}(f_K)]'$, and $\hat{\Omega}$ is the variance-covariance matrix of the factors,

$$\hat{\Omega} = \frac{1}{T}\sum_{t=1}^{T}\left[f_t - \hat{E}(f_t)\right]\left[f_t - \hat{E}(f_t)\right]'.$$

- **The algorithm for implementing the statistic is**:
  1. For each asset, run TS regressions of the asset returns on the common factors.
  2. Retain the estimated regression intercepts $\hat{\alpha} = [\hat{\alpha}_1, \hat{\alpha}_2, \ldots, \hat{\alpha}_N]'$ and the estimated residuals $\epsilon_t = [\epsilon_t^1, \epsilon_t^2, \ldots, \epsilon_t^N]'$ for $t = [1, 2, \ldots, T]$.
  3. Compute the residual covariance matrix $\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T} \epsilon_t \epsilon_t'$.
  4. Compute the vector of sample means, $\hat{E}(f) = [\hat{E}(f_1), \hat{E}(f_2), ..., \hat{E}(f_K)]'$. Compute the variance-covariance matrix of the factors:

$$\hat{\Omega} = \frac{1}{T}\sum_{t=1}^{T}\left[f_t - \hat{E}(f_t)\right]\left[f_t - \hat{E}(f_t)\right]' \tag{54}$$

  5. Compute J_3 as in Equation 53.

2.1.3 Economics of the time-series test statistics

- **GRS test whether the factor f is *ex-ante* mean-variance efficient**—whether it is on the mean-variance [MV] frontier using *population* moments—after accounting for sampling error.
- Even if f is on the *ex-ante* MV frontier, other returns will outperform it in sample due to luck, so f will usually be inside the ex-post MV frontier—i.e. the frontier drawn using *sample* moments, **still it should not be too far**.
- GRS show that their test statistic can be expressed in terms of **how far inside the *ex-post* frontier the reference return is**

$$J_2 = \frac{T-N-1}{N}\left[\frac{(\mu_q/\sigma_q)^2 - \left(\hat{E}(f)/(\hat{\sigma}(f)\right)^2}{1+\left(\hat{E}(f)/(\hat{\sigma}(f)\right)^2}\right], \tag{55}$$

where $(\mu_q/\sigma_q)^2$ is the Sharpe ratio of the *ex-post* tangency portfolio formed from the test assets *plus* the factor f.

2.1.4 Testing whether a factor is redundant

- **You may be tempted** to test whether 'one of the factors is priced' by testing whether its risk premium is significant:

$$\frac{\hat{\lambda}_i}{\sigma(\hat{\lambda}_i)} \sim t_{T-1} \tag{56}$$

- **However, this is *incorrect* to do in the time-series tests.**
- **Whether its risk premium is significant or not does not imply *whether it can price other assets*.**
- For example, consider the FF93 factors. The risk premium for SMB is small, i.e. $\hat{\lambda}_{SMB} \approx 0$. Is SMB redundant? Should we drop it?
- Consider two alternatives:
  1. $\hat{\lambda}_{SMB} > 0$, but it is spurious—it does not price other assets.
  2. $\hat{\lambda}_{SMB} = 0$, but it may be important—it may price other assets.

1. $\hat{\lambda}_{SMB} > 0$, **but it is spurious — it does not price other assets.**
   - Let $SMB_t = \frac{1}{2}MKT_t + \frac{1}{2}HML_t$, and MKT and HML are important.
   - Naturally SMB commands a risk premium for its exposure to MKT and HML: $E(SMB) > 0$, so $\hat{\lambda}_{SMB} > 0$.
   - But does it mean SMB carries information beyond MKT and HML?
   - The right way to test this is to simply regress SMB on MKT and HML:

$$SMB_t = \alpha_s + b_s MKT_t + h_s HML_t + e_t \tag{57}$$

   to see if $\alpha_s = 0$.
   $\longmapsto$ **If and only if $\alpha_s = 0$, we can drop SMB — it is priced by the other assets.**

2. $\hat{\lambda}_{SMB} = 0$, **but it may be important — it may price other assets.**
   - If $\alpha_s \neq 0$ in (57), do not drop SMB, it is not priced by the other factors — it *may* carry additional information to explain the prices of other assets.

- **We need to test whether** *SMB prices other assets*, **not whether it has a positive risk premium.**
  Example: Apple may have a positive risk premium, because it has a MKT beta of about 1.00, but it doesn't mean Apple can be used to price other securities.

2.1.5 Summary of the time-series asset-pricing tests

- The central question in the time-series framework is **whether the factor is ex ante mean-variance efficient** (or, in multi-factor models, whether some combination of the factors is on the ex ante efficient frontier).
- **Time-series test statistics indicate how far away is the tested factor** (or a combination of factors) **from the efficient frontier**.
- **Testing the joint equality of the pricing errors to zero amounts to testing whether the portfolio-based factor is ex-ante mean variance efficient** — whether it is on the mean-variance efficient frontier based on *population* moments.

2.1.6 When are time-series asset-pricing tests appropriate?

- **The factors are returns**.
  $\longmapsto$ Otherwise, the alpha is hard to interpret – not a 'pricing error'.
  $\longmapsto$ *Consumption growth* is a good example of a non-portfolio-based factor.
- **A long time-series and a small cross-section of test assets**.
  $\longmapsto$ Otherwise, the variance-covariance matrices cannot be inverted.
- **The test assets are portfolios rather than individual securities**.
  $\longmapsto$ The test statistic requires inverting variance covariance matrices with *non-missing values*.
  $\longmapsto$ We need a *balanced panel* – each asset' time-series needs to have the same length.
  $\longmapsto$ There are not that many stocks with long time-series. Picking just those with long time-series introduces survivorship bias and produces an unrepresentative sample.

2.2 Cross-Sectional Regressions [CSR]

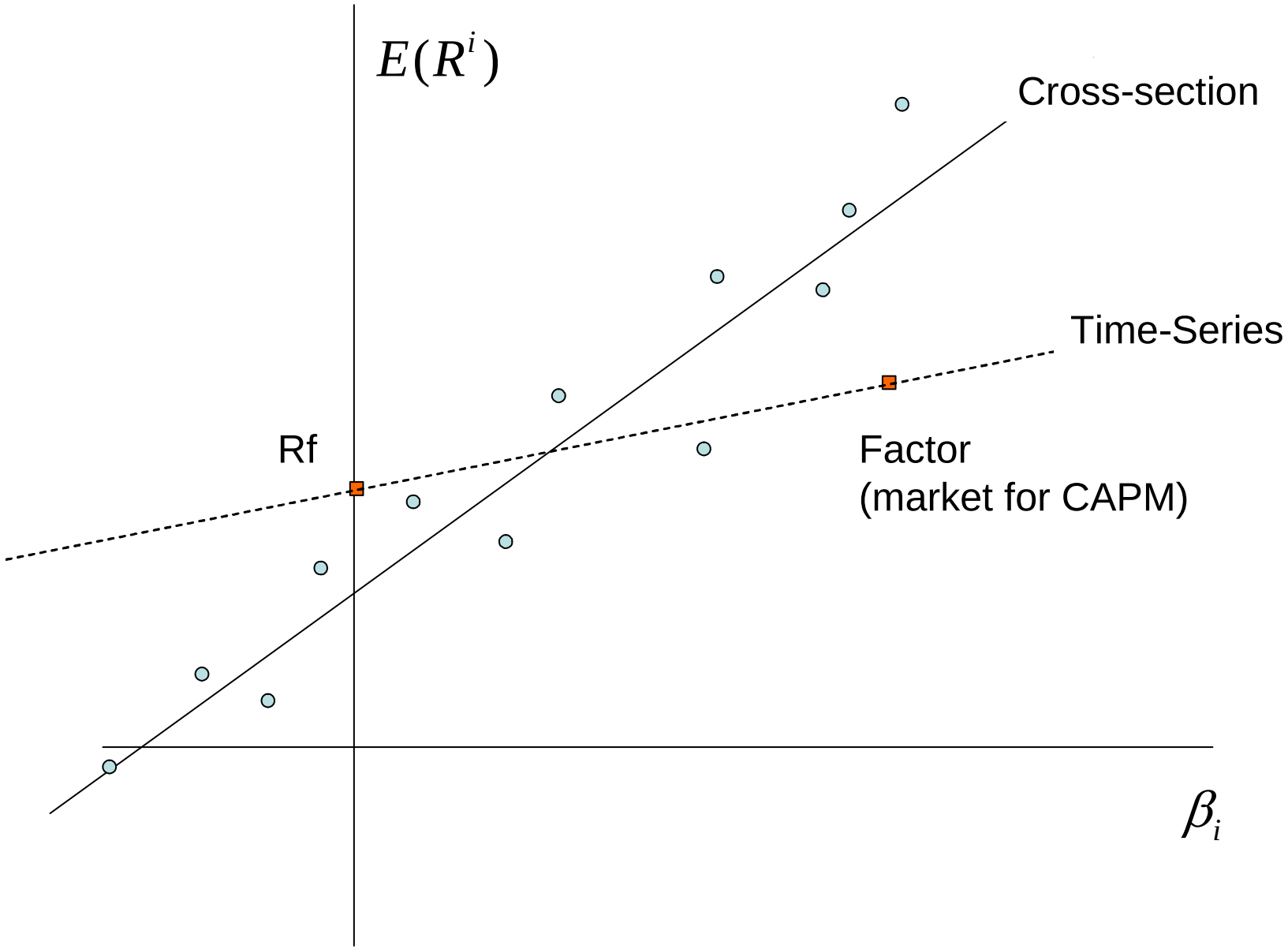
- **Can be implemented whether or not the factor is a return spread.**
- The central question in the cross-sectional framework is
  Why do average returns vary *across* assets?
- The asset pricing models say that
  Expected returns should be high if the asset has high betas.
- So plot the sample *average* excess returns on the estimated betas. **(Matlab)**

  $\longmapsto$ Even if the model is correct, this plot will not work out perfectly well because of sampling errors.

- **The idea is to run a cross-sectional regression to fit a line through the scatterplot of average returns on estimated betas.**

- The cross-sectional (CS) tests involve:
  1. **Examining whether the 'pricing errors' are 'too' large**. The 'pricing errors' are the *error terms* in the cross-sectional regressions. (*not the intercepts in the TSR*, as we no longer claim $\hat{\lambda} \equiv \hat{E}(f)$).
  2. **Examining whether the factor is priced** – whether the slope coefficient, λ, in the CS regression of average returns on betas is significant.

TS vs CSR


$E(R^i)$
Cross-section
Time-Series
Rf
Factor
(market for CAPM)
β_i

- Note that the average returns and betas are the same for both TSR and CSR — betas are estimated from the TSR in both cases.
- TS: The risk premium is *fixed* (at the sample mean of the factor): $\lambda = \hat{E}(f_t)$.
- CS: The risk premium is *estimated* as the slope coefficient in the CSR (eq.59).

- **Two regression steps** are at the heart of the cross-sectional approach:
  1. *Step 1* [TSR]: **Estimate betas from time-series regression** of excess returns on some pre-specified factors
  $$r_{i,t} = a_i + \beta_i' f_t + \epsilon_{i,t}. \tag{58}$$
  2. *Step 2* [CRS]: **Run a cross-section regression** of *average* returns on the betas
  $$\bar{r}_i = \lambda' \hat{\beta}_i + \alpha_i. \tag{59}$$

NOTE THAT:

- λ**, the CSR coefficient, is the estimate of the 'risk premium'** (it no longer is the sample mean of the factor, $\hat{\lambda} \neq E(f)$)
- α_i**, the error term in the CRS, is the 'pricing error'** (the pricing errors are no longer the a_i because $\hat{\lambda} \neq E(f)$

  $\bar{r} = [\bar{r}_1, \bar{r}_2, \ldots, \bar{r}_N]'$, are the sample average asset returns

  $\beta = [\beta_1, \beta_2, \ldots, \beta_N]'$ are estimated in the first pass and use as regressors in the second pass

- **Intercept or no intercept in the cross-sectional regression?**
  - You should include an intercept in the time-series regression (eq.58).
  - But you *can* run the cross-sectional regression (eq.59 or eq.60) with or without intercept:

$$\bar{r}_i = \gamma + \lambda' \hat{\beta}_i + \alpha_i. \tag{60}$$

    i.e. force $\gamma = 0$ (no intercept) or allow $\gamma \neq 0$ (with intercept).
  - Theory says that the constant, or the zero-beta excess return, should be zero.
    $\longmapsto$ You can include a constant and see if it turns out to be small.
  - It's a trade-off between *efficiency* (*no intercept*, imposing the null gets more efficient estimates) and *robustness* (*with intercept*).

- For now, let's assume that there is a single factor and no intercept.

The OLS cross-sectional estimates are

$$\hat{\lambda} = (\beta'\beta)^{-1}\beta'\bar{r}, \tag{61}$$

$$\hat{\alpha} = \bar{r} - \hat{\lambda}\beta. \tag{62}$$

- You *cannot* use the standard erros from the OLS CRS as $\sigma(\hat{\lambda})$, because the errors α_i are correlated.

- Since the α_i are just the time-series average of the true ϵ_{it} shocks (not the TSR residuals which are always mean zero), the errors in the CSR have covariance matrix $E(\alpha\alpha') = \frac{1}{T}\Sigma = \frac{1}{T}E(\epsilon_t\epsilon_t')$.

The standard errors of $\hat{\lambda}$ and $\hat{\alpha}$ are:

$$\sigma^2(\hat{\lambda}) = \frac{1}{T}(\beta'\beta)^{-1}\beta'\Sigma\beta(\beta'\beta)^{-1}, \tag{63}$$

$$\text{cov}(\hat{\alpha}) = \frac{1}{T}\left(I - \beta(\beta'\beta)^{-1}\beta'\right)\Sigma\left(I - \beta(\beta'\beta)^{-1}\beta'\right). \tag{64}$$

Test whether all pricing errors are zero with the statistic:

$$\hat{\alpha}' cov(\hat{\alpha})^{-1} \hat{\alpha} \sim \chi^2_{N-K}. \tag{65}$$

Since $cov(\hat{\alpha})$ is singular, we have to use a generalized inverse (pinv in matlab).

We could also test whether a factor is priced

$$\frac{\hat{\lambda}}{\sigma(\hat{\lambda})} \sim t_{N-K} \tag{66}$$

2.2.1 Errors-in-variables [EIV]

- Since β in the CSR are themselves estimated in the TSR,

  we have an EIV problem—β is not known.

- Shanken (1992) corrects the CSR estimates to account for the EIV problem.

The CSR standard erros with the Shanken (1992) correction are:

$$\sigma^2(\hat{\lambda}) = \frac{1}{T}\left[(\beta'\beta)^{-1}\beta'\Sigma\beta(\beta'\beta)^{-1}(1+\lambda'\Sigma_f^{-1}\lambda)+\Sigma_f\right], \tag{67}$$

$$\text{cov}(\hat{\alpha}) = \frac{1}{T}\left(I-\beta(\beta'\beta)^{-1}\beta'\right)\Sigma\left(I-\beta(\beta'\beta)^{-1}\beta'\right)(1+\lambda'\Sigma_f^{-1}\lambda) \tag{68}$$

Then test

$$\hat{\alpha}'cov(\hat{\alpha})^{-1}\hat{\alpha} \sim \chi^2_{N-K-1} \tag{69}$$

- The *additive term,* Σ_f, is the variance-covariance matrix of the factors.
- If the factors are returns, the *multiplicative term,* $\Upsilon = 1+\lambda'\Sigma_f^{-1}\lambda$, is the squared Sharpe ratio of the tangency portfolio of the factors.

- How important is the Shanken correction?
  1. **The multiplicative term** (affects both $\sigma(\hat{\lambda})$ and $\sigma(\hat{\alpha})$).
     - **In annual data**: In the CAPM, $\lambda = Er_m$, so $\lambda^2/\sigma^2(r_m) = (0.08/0.16)^2 = 0.25$. So the correction implies that the variance of the risk premium estimate is 25% larger, i.e. $\Upsilon = 1.25$. This is a **very large correction**.
     - **In monthly data**: Mean and variances both scale with horizon, i.e. λ and σ^2 are 12 times smaller. Hence, $\lambda^2/\sigma^2(r_m) \approx 0.25/12 \approx 0.02$. So $\Upsilon \approx 1.02$, a 2% correction – **a minor correction**.
  2. **The additive term** (affects only $\sigma(\hat{\lambda})$)

     is an important correction.
- Note that the Shanken correction makes it more difficult to reject the model.
- Note that for now we are still dealing with *balanced panels* (no missing observtions) and a small number of test assets. This is because we're still multiplying matrices and need variance-covariance matrices.
- Balanced panels will not be required for the Fama-MacBeth method we study next.

2.3 Fama and MacBeth Method [FM]

- Fama and MacBeth (1973) propose an alternative procedure for running cross-sectional regressions, which is very widely used in practice.
- The FM approach is an extension to the CRS approach.
- The first step is identical to the one described above.
- In the second step, instead of estimating a single cross-sectional regression with the sample averages on the estimated betas, FM run a cross-sectional regression at *each time period*
- **The Fama-MacBeth two-pass approach** is as follows:
  1. **I Pass:**: **For each asset, estimate betas from a TSR** of excess returns on some pre-specified factors:

$$r_{i,t} = a_i + \beta_i' f_t + \epsilon_{i,t}. \tag{70}$$

  2. **II Pass:**: **Each month, run a cross-section regression** of excess returns on the estimated betas:

$$r_{i,t} = \lambda_{0,t} + \lambda_t' \hat{\beta}_i + \alpha_{i,t}. \tag{71}$$

- **Each period** t, the estimates of $\lambda_{0,t}$, λ_t, and $\alpha_{i,t}$ are the **OLS estimates**.

The estimates of λ (as well as λ_0) and α_i are the *averages across time*:

$$\hat{\lambda} = \frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_t, \tag{72}$$

$$\hat{\alpha}_i = \frac{1}{T}\sum_{t=1}^{T}\hat{\alpha}_{i,t}. \tag{73}$$

The standard errors are based on the sample variance of the cross-sectional regression estimates and use the fact that $\sigma^2(\overline{x}) = \sigma^2(x)/T$:

$$\sigma^2(\hat{\lambda}) = \frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\lambda}_t - \hat{\lambda}\right)^2, \tag{74}$$

$$\sigma^2(\hat{\alpha}_i) = \frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\alpha}_{i,t} - \hat{\alpha}_i\right)^2. \tag{75}$$

- The FM procedure uses the variation of the statistic *over time* to deduce its variation across samples.

- The variance-covariance matrix of the sample pricing errors is

$$cov(\hat{\alpha}) = \frac{1}{T^2}\sum_{t=1}^{T}(\hat{\alpha}_t - \hat{\alpha})(\hat{\alpha}_t - \hat{\alpha})'. \tag{76}$$

- Then we can test whether all pricing errors are zero using the test statistic:

$$\hat{\alpha}' cov(\hat{\alpha})^{-1}\hat{\alpha} \sim \chi^2_{N-K}. \tag{77}$$

Test whether all pricing errors are zero with the statistic:

$$\hat{\alpha}' cov(\hat{\alpha})^{-1}\hat{\alpha} \sim \chi^2_{N-K}. \tag{78}$$

We could also test whether a factor is priced

$$\frac{\hat{\lambda}}{\sigma(\hat{\lambda})} \sim t_{T-1} \tag{79}$$

- The test in equation (79) is the most commonly used.

2.3.1 EIV in the Fama-MacBeth approach

- The EIV problem is still present in FM, because betas are still estimated.
- Fama and MacBeth (1973) propose forming portfolios to resolve the problem.
- Shanken (1992) propose the following correction to the standard errors involving a quadratic term: $c = \lambda' \Sigma_f^{-1} \lambda$

The Shanken (1992) correction

$$T\sigma^2_{EIV}(\hat{\lambda}) = (1+c)[T\sigma^2(\hat{\lambda}) - \Sigma_f] + \Sigma_f \tag{80}$$

where $\sigma^2(\hat{\lambda})$ is the Fama-MacBeth variance given in eq.(74).

- Many researchers have argued that the Shanken correction **matters little in practice when the factors are portfolio excess returns** (i.e. traded-portfolios).
- Kan, Robotti, and Shanken (2013) show that **substantial error adjustments can be obtained with non-traded factors.**

2.3.2 Advantages of Fama-MacBeth approach

- *It can accomodate unbalanced panels.* Each month t, you can use a different number of stocks – those that exist at time t. The distribution of the risk premium does not depend on the number of stocks.
- *Betas can be time-varying.*
  - Fama and MacBeth (1973) use *60-month rolling windows* to estimate the betas for each period. Specifically, the betas used in period t are estimated using TSR over months $t-61$ to $t-1$. The idea is that investors condition on information they have up to period t. However, Fama and French (1992) argue that using the full-sample betas does not yield different results.
  - Betas can also be *conditioned on instruments* (more on this later).
  - Betas can *vary with firm characteristics* (more on this later).
- It is possible that *autocorrelation in returns* (negligible at monthly frequencies) leads to autocorrelation in risk premium estimates. This is easily corrected using Newey and West (1987) type corrections.
- *It is easy to implement.*

2.4 Portfolios or Individual Securities in Asset Pricing Tests

- Using **portfolios** helps with estimation issues:
  - To mitigate the errors-in-variables problem due to using estimated betas as independent variables in the two-step procedure.
  - To allow estimation of the residual covariance matrix in one-step procedures.
- Using **single stocks** to test asset pricing models is more robust because:
  - It avoids data-snooping biases that are inherent in portfolio-based approaches, as noted by Lo and MacKinlay (1990).
  - It is also robust to the sensitivity of asset pricing tests to the portfolio grouping procedure.
  - Portfolio averaging conceals possibly relevant security characteristics.
  - It allows to study the impact of firm-level characteristics on the cross-section of stock returns.

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